

PROJECT SYNOPSIS

for

Restaurant Website & Online Ordering System

A Web-Based Platform for Menu Browsing, Table Reservation, Online Food Ordering and Order Tracking

Submitted To	GLA University
Course / Program	B.Tech – 3rd Semester
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1. Introduction

Restaurants today increasingly rely on digital platforms to reach customers, streamline table bookings, and manage food orders efficiently. Manual, phone-based or walk-in-only order and reservation handling is slow, error-prone, and offers no way for customers to browse the menu, check table availability, or track their order in real time before or during their visit.

The Restaurant Website & Online Ordering System (RWOOS) is a web-based application proposed to address this gap for a single restaurant. It allows customers to browse a digital menu, reserve a table, place an order for dine-in, takeaway or delivery, and track the status of that order — while giving restaurant staff and management a simple dashboard to handle incoming orders, reservations, the menu, and daily reporting.

2. Objectives of the Project

- To design and develop a responsive restaurant website that allows customers to browse the menu by category and place orders online.
- To implement a table reservation system that prevents double-booking of the same table and time slot.
- To provide customers with real-time order status tracking, from order placement to completion.
- To build a staff dashboard for managing incoming orders and daily table reservations efficiently.
- To build an administrator panel for managing the menu, categories, tables, staff accounts and basic sales reports.
- To apply secure authentication and role-based access control across customer, staff and admin roles.

3. Scope of the Project

The system is scoped for a single restaurant / single branch and is intended to be built and demonstrated within one college semester by a small student team. It covers menu browsing, table reservation, order placement/tracking, and basic staff and admin dashboards with reporting.

The following are explicitly out of scope for the current version and are proposed only as future enhancements: online payment gateway integration, SMS notifications, loyalty/rewards programs, third-party delivery-partner integration, and multi-branch/franchise management.

4. Existing System vs. Proposed System

4.1 Existing System

- Orders and table bookings are typically taken over the phone or in person, which is time-consuming and error-prone.

- Customers have no way to view the live menu, prices or table availability before visiting or calling.
- There is no organized way for staff to track order status or for management to view sales trends.

4.2 Proposed System

- A self-service website where customers can browse the menu, reserve tables, and place orders online at any time.
- Automatic prevention of double-booked tables and real-time order status visible to the customer.
- A centralized staff dashboard for order and reservation management, and an admin panel for configuration and reporting.

5. Modules of the Proposed System

Module	Description
User Registration & Authentication	Secure sign-up/login for customers, staff and admin with role-based access (JWT-based sessions).
Customer Portal	Menu browsing, cart & checkout, table reservation, order tracking, profile, and item ratings/reviews.
Staff Portal	Live incoming-order board, order status updates, and daily table/reservation management.
Administrator Portal	Menu, category, table and staff management; order/reservation oversight; sales and workload reports.
Notifications	Automated email confirmations for orders, reservations, cancellations and status changes.

6. Technology Stack

Layer	Technology
Frontend	React.js, React Router, Axios, Bootstrap / Tailwind CSS
Backend	Node.js, Express.js, JWT, bcrypt, Nodemailer
Database	MongoDB with Mongoose (MERN stack)
Version Control	Git & GitHub
Deployment (suggested)	Vercel/Netlify (frontend), Render/Railway (backend), MongoDB Atlas (database)

7. Hardware & Software Requirements

7.1 Hardware Requirements

- A standard desktop/laptop for development (minimum 4 GB RAM, dual-core processor).
- Client devices: any desktop, laptop, tablet or smartphone with an internet connection.
- No specialized restaurant hardware (e.g. POS terminals) is required for the academic version.

7.2 Software Requirements

- Operating System: Windows / Linux / macOS.
- Node.js runtime, MongoDB (local or Atlas cloud), and a modern code editor (e.g. VS Code).
- A modern web browser (Chrome, Firefox, Edge or Safari) for testing and demonstration.

8. Feasibility Study

8.1 Technical Feasibility

The project uses widely documented, free and open-source technologies (MongoDB, Express.js, React.js, Node.js) that are well within the skill range of a B.Tech student team and are supported by extensive community documentation.

8.2 Economic Feasibility

All proposed tools, frameworks and hosting tiers (Vercel/Netlify, Render/Railway, MongoDB Atlas free tier) are free for student/demo-scale use, keeping the project cost effectively at zero.

8.3 Operational Feasibility

The interface is designed to be simple enough for a first-time customer to place an order in a few clicks, and the staff/admin dashboards require only basic computer literacy, making the system practical to operate and demonstrate.

9. Expected Outcome

A fully functional, responsive restaurant website supporting menu browsing, table reservation, online order placement (dine-in/takeaway/delivery) and real-time order tracking, backed by a working staff dashboard for order/reservation management and an admin panel for menu, table, staff and report management — ready for faculty evaluation and live demonstration.

10. Future Scope

- Integration of an online payment gateway for order checkout.
- SMS notifications alongside email for order and reservation updates.

- A loyalty/rewards program for repeat customers.
- Integration with third-party delivery partners.
- Support for multi-branch/franchise restaurant management.
- AI-based menu recommendations based on customer order history.

11. References

- React.js, Node.js, Express.js and MongoDB official documentation.
- IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications.
- Project SRS document prepared for the Restaurant Website & Online Ordering System.